STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-1-MORPHISMS-PRESHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 369-378 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 380-437, the complete author proof (56 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_algebraic-stacks-sheaves.json (pb=2136, stmt 369-378, proof 380-437).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 002L: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 437–437. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$. Then $(g \circ f)^p = f^p \circ g^p$ and there is a canonical isomorphism ${}_p(g \circ f) \to {}_pg \circ {}_pf$ compatible with adjointness of $(f^p, {}_pf)$, $(g^p, {}_pg)$, and $((g \circ f)^p, {}_p(g \circ f))$.*

Proof. Let $\mathcal{H}$ be a presheaf on $\mathcal{Z}$. Then $(g \circ f)^p\mathcal{H} = f^p(g^p\mathcal{H})$ is given by the equalities

$$(g \circ f)^p\mathcal{H}(x) = \mathcal{H}((g \circ f)(x)) = \mathcal{H}(g(f(x))) = f^p(g^p\mathcal{H})(x).$$

We omit the verification that this is compatible with restriction maps.

Next, we define the transformation ${}_p(g \circ f) \to {}_pg \circ {}_pf$. Let $\mathcal{F}$ be a presheaf on $\mathcal{X}$. If z is an object of $\mathcal{Z}$ then we get a category $\mathcal{J}$ of quadruples $(x, f(x) \to y, y, g(y) \to z)$ and a category $\mathcal{I}$ of pairs $(x, g(f(x)) \to z)$. There is a canonical functor $\mathcal{J} \to \mathcal{I}$ sending the object $(x, \alpha : f(x) \to y, y, \beta : g(y) \to z)$ to $(x, \beta \circ f(\alpha) : g(f(x)) \to z)$.

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This gives the arrow in

$$\begin{aligned}
({}_p(g \circ f)\mathcal{F})(z) &= \lim_{g(f(x)) \to z} \mathcal{F}(x) \\
&= \lim_{\mathcal{I}} \mathcal{F} \\
&\to \lim_{\mathcal{J}} \mathcal{F} \\
&= \lim_{g(y) \to z} \left(\lim_{f(x) \to y} \mathcal{F}(x) \right) \\
&= ({}_pg \circ {}_pf\mathcal{F})(x)
\end{aligned}$$

by Categories, Lemma 002L. We omit the verification that this is compatible with restriction maps. An alternative to this direct construction is to define ${}_p(g \circ f) \cong {}_pg \circ {}_pf$ as the unique map compatible with the adjointness properties. This also has the advantage that one does not need to prove the compatibility.

Compatibility with adjointness of $(f^p, {}_pf)$, $(g^p, {}_pg)$, and $((g \circ f)^p, {}_p(g \circ f))$ means that given presheaves $\mathcal{H}$ and $\mathcal{F}$ as above we have a commutative diagram

$$\begin{array}{ccccc}
\mathrm{Mor}_{PSh(\mathcal{X})}(f^p g^p \mathcal{H}, \mathcal{F}) & = & \mathrm{Mor}_{PSh(\mathcal{Y})}(g^p \mathcal{H}, {}_pf\mathcal{F}) & = & \mathrm{Mor}_{PSh(\mathcal{Y})}(\mathcal{H}, {}_pg{}_pf\mathcal{F}) \\
\| & & & & \uparrow \\
\mathrm{Mor}_{PSh(\mathcal{X})}((g \circ f)^p \mathcal{G}, \mathcal{F}) & & = & & \mathrm{Mor}_{PSh(\mathcal{Y})}(\mathcal{G}, {}_p(g \circ f)\mathcal{F})
\end{array}$$

Proof omitted. □